11 Chemistry
Unit 2 Topic Test 6: Redox

Details	
Student Name:	Teacher Name:
Date Set: ____/____/26	Due Date: ____/____/26

Curriculum Assessed

On completion of Unit 2 the student should be able to measure and analyse chemicals in water, including redox reactions. This topic test assesses oxidising and reducing agents, oxidation numbers, balanced half-equations and overall equations (including in acidic conditions), the reactivity series and metal displacement using the electrochemical series, corrosion and its prevention, and simple primary galvanic cells.

Task Overview

Section A – Multiple Choice: 10 questions, 1 mark each (10 marks). Answer on the separate multiple-choice answer sheet.
Section B – Short Answer: 4 questions (30 marks). Answer in this booklet.

Conditions

- Test conditions (revision topic test)
- VCE Chemistry Data Book provided
- Permitted materials: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator (no CAS).
- Not permitted: notes of any kind, blank paper, white-out, mobile phones, smart watches or any other electronic device.
- 46 mins writing time (about 1.15 minutes per mark)

Marking

- 40 marks
- Marked according to teacher-made marking guide.
- Show all working in calculation questions; give final answers with units and an appropriate number of significant figures. Equations must be balanced and include states.

A+	A	B+	B	C+	C	D+	D	E+	E	NS
90-100%	80-89%	75-79%	70-74%	65-69%	60-64%	55-59%	50-54%	45-49%	40-44%	<40%

Total Marks:		Percentage:		Letter Grade:	

SECTION A – Multiple-choice questions (10 marks)

Answer **all** questions on the separate multiple-choice answer sheet by filling in the letter of the best response. Each question is worth 1 mark. No marks are deducted for incorrect answers. Do **not** write your Section A answers in this booklet.

Question 1

Which one of the following correctly describes a reducing agent?

	What happens to electrons	What happens to the reducing agent
A.	it accepts electrons	it is reduced
B.	it donates electrons	it is oxidised
C.	it donates electrons	it is reduced
D.	it accepts electrons	it is oxidised

Question 2

Using the electrochemical series in the Data Book, which one of the following metals is the strongest reducing agent?

A. nickel, Ni

B. lead, Pb

C. aluminium, Al

D. tin, Sn

Question 3

Chromate ions, CrO_4^{2-}, give some paints and pigments a yellow colour.

The oxidation number of chromium in CrO_4^{2-} is

A. +6

B. +8

C. +10

D. −2

Question 4

Magnesium, Mg, keeps burning in carbon dioxide, CO_2, so carbon dioxide fire extinguishers must not be used on magnesium fires.

$$2Mg(s) + CO_2(g) \rightarrow 2MgO(s) + C(s)$$

The number of electrons transferred for each molecule of CO_2 that reacts is

A. 1

B. 2

C. 4

D. 8

Question 5

Consider the following half-equation.

$$VO_2^+(aq) + 2H^+(aq) + e^- \rightarrow VO^{2+}(aq) + H_2O(l)$$

Which one of the following statements about this half-equation is correct?

A. Vanadium is reduced from +5 to +4.

B. Vanadium is oxidised from +4 to +5.

C. H^+ is the species that is reduced.

D. The half-equation is not balanced for charge.

Use the following information to answer Questions 6 and 7.

Copper, Cu, roofs and domes slowly turn green as a layer of basic copper carbonate, $Cu_2(OH)_2CO_3$, forms on the surface.

$$2Cu(s) + O_2(g) + H_2O(l) + CO_2(g) \rightarrow Cu_2(OH)_2CO_3(s)$$

Question 6

In this reaction, the oxidation number of copper changes from

A. 0 to +1

B. +2 to 0

C. 0 to +4

D. 0 to +2

Question 7

Which species is the oxidising agent in this reaction?

A. Cu

B. O_2

C. H_2O

D. CO_2

Question 8

Aluminium foil, Al, is placed in excess copper(II) chloride solution, $CuCl_2(aq)$. Copper metal, Cu, forms and aluminium ions, Al^{3+}, go into solution.

If 0.540 g of aluminium reacts completely, the mass of copper produced is closest to

A. 1.27 g

B. 0.847 g

C. 3.81 g

D. 1.91 g

Question 9

Consider the following statements about a simple primary galvanic cell made from two metal half-cells joined by a salt bridge.

I. Electrons flow through the external circuit from the anode to the cathode.

II. The mass of the anode decreases as the cell operates.

III. The cathode is the negative electrode.

Which of these statements are correct?

A. I only
B. I and II only
C. II and III only
D. I, II and III

Question 10

For iron, Fe, to rust, which substances must be present?

A. oxygen only
B. water only
C. both oxygen and water
D. both nitrogen and water

END OF SECTION A

SECTION B – Short-answer questions (30 marks)

Answer **all** questions in the spaces provided. Show all working in calculation questions. Give final answers with units and an appropriate number of significant figures. Equations must be balanced and include states, for example $H_2(g)$, NaCl(s).

Question 1 (8 marks)

In a home aquarium, fish waste releases ammonium ions, NH_4^+, which are toxic to fish. Bacteria living in the filter use oxygen, O_2, dissolved in the water to oxidise the ammonium ions to nitrite ions, NO_2^-. Other bacteria then oxidise the nitrite ions to nitrate ions, NO_3^-, which are much less harmful to fish. Assume that all of these reactions take place in acidic conditions.

a) State the oxidation number of nitrogen in NH_4^+.

1 mark

b) State the oxidation number of nitrogen in NO_2^-.

1 mark

c) Write the balanced half-equation for the oxidation of $NH_4^+(aq)$ to $NO_2^-(aq)$ in acidic conditions. Include states.

2 marks

d) Identify the oxidising agent in the overall reaction in which NH_4^+ is oxidised to NO_2^-.

1 mark

e) The reduction half-equation is $O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$. Use this and your answer to part c) to write the balanced overall equation, including states.

2 marks

f) If an aquarium filter runs short of oxygen, other bacteria reduce nitrate ions, $NO_3^-(aq)$, back to nitrite ions, $NO_2^-(aq)$. Write the balanced half-equation for this reduction in acidic conditions, including states.

1 mark

Question 2 (9 marks)

Silver, Ag, cutlery darkens over time as a thin layer of black silver sulfide, Ag_2S, forms on its surface. One cleaning method is to line a glass dish with aluminium foil, Al, place the cutlery on the foil so that it touches it, and cover everything with hot sodium hydrogen carbonate solution, $NaHCO_3(aq)$. The black layer slowly disappears and shiny silver is left.

Assume that the only products are silver, Ag(s), aluminium ions, $Al^{3+}(aq)$, and sulfide ions, $S^{2-}(aq)$. The simplified half-equation for the reaction of the tarnish is

$$Ag_2S(s) + 2e^- \rightarrow 2Ag(s) + S^{2-}(aq)$$

a) Write the half-equation for the oxidation of aluminium to aluminium ions, Al^{3+}, including states.

1 mark

b) Write the balanced overall equation for the reaction, including states.

2 marks

c) Identify the reducing agent in this reaction.

1 mark

d) The cutlery must be touching the aluminium foil for the tarnish to be removed. Explain why, in terms of electron transfer.

2 marks

e) A tarnished bowl has 0.124 g of Ag_2S on its surface. Calculate the mass of aluminium that is oxidised when all of this Ag_2S is converted to silver.

2 marks

f) State why the foil and cutlery must be covered by a solution of ions, rather than by pure water.

1 mark

Question 3 (8 marks)

Rohan built the galvanic cell shown in Figure 1. A strip of magnesium, Mg, stood in 1.0 M magnesium nitrate solution, $Mg(NO_3)_2(aq)$, and a strip of nickel, Ni, stood in 1.0 M nickel(II) nitrate solution, $Ni(NO_3)_2(aq)$. The salt bridge was a strip of filter paper soaked in sodium sulfate solution, $Na_2SO_4(aq)$. A voltmeter was connected between the two electrodes.

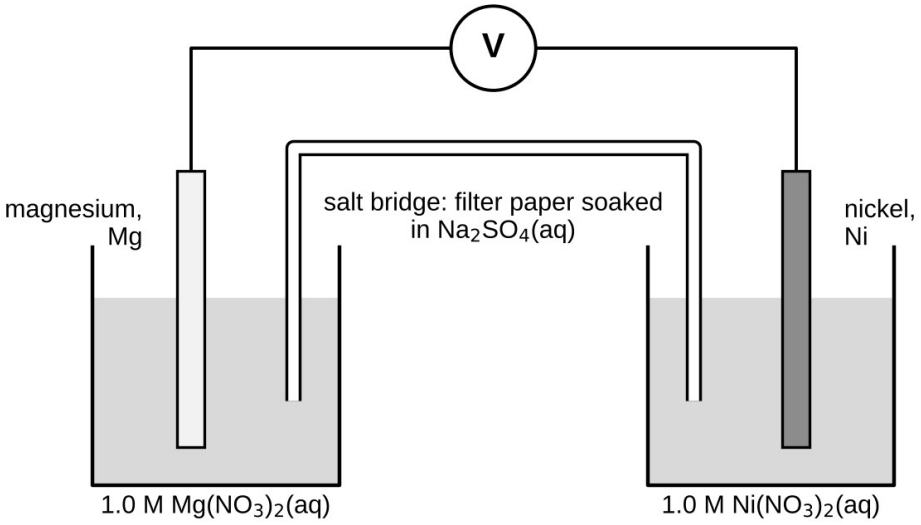


Figure 1 A magnesium–nickel galvanic cell

a) Identify the electrode at which oxidation occurs.

1 mark

b) Write the half-equation for the reaction at the anode and the half-equation for the reaction at the cathode, including states. Label each one.

2 marks

c) State the polarity (positive or negative) of the nickel electrode.

1 mark

d) Explain which half-cell the sulfate ions, SO_4^{2-}, in the salt bridge move towards as the cell operates.

2 marks

e) Rohan's friend suggests a simpler set-up: put the magnesium strip straight into the nickel(II) nitrate solution, with no salt bridge. Explain why this would not produce an electric current through the voltmeter.

2 marks

Question 4 (5 marks)

Ava pressed four pieces of iron wire, Fe, into a dish of warm agar gel. The gel contains a chemical that turns blue wherever iron(II) ions, Fe^{2+}, form. Her results after two days are shown below.

Piece	Treatment	Observation after two days
A	bare iron wire	blue patches near both ends
B	strip of magnesium, Mg, wrapped tightly around the middle	no blue colour anywhere on the wire
C	painted all over, then scratched through the paint along the middle	blue colour only along the scratch
D	completely coated with clear lacquer (a hard, waterproof varnish)	no blue colour

a) State what the blue colour around piece A shows has happened to the iron.

1 mark

b) State how the lacquer prevents piece D from corroding.

1 mark

c) Write the half-equation for the reaction of the magnesium strip on piece B, including states.

1 mark

d) Explain, with reference to the electrochemical series, why no blue colour formed anywhere on piece B, but blue colour formed along the scratch on piece C.

2 marks

END OF SECTION B

Additional space

Clearly number any response written here.

END OF TEST

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SECTION A – MULTIPLE-CHOICE ANSWER SHEET

Student name: _____ Teacher: _____

Shade or circle ONE letter for each question. Each question is worth 1 mark. No marks are deducted for incorrect answers.

Question	A	B	C	D
1	(A)	(B)	(C)	(D)
2	(A)	(B)	(C)	(D)
3	(A)	(B)	(C)	(D)
4	(A)	(B)	(C)	(D)
5	(A)	(B)	(C)	(D)
6	(A)	(B)	(C)	(D)
7	(A)	(B)	(C)	(D)
8	(A)	(B)	(C)	(D)
9	(A)	(B)	(C)	(D)
10	(A)	(B)	(C)	(D)

Section A score: _____ / 10